



Learning Analytics in Online Learning Environment: A Systematic Review on the Focuses and the Types of Student-Related Analytics Data

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Abstract

The application of learning analytics in an online learning environment is increasing among researchers in educational fields because it can assist in providing standard and measurable decision making about student success. In this regard, there is a need for the online learning society and practitioners to be informed about how learning analytics is applied in the online learning environment. Therefore, this systematic review article aims to offer the reader essential details regarding the practical usage of learning analytics techniques in online learning environments to improve the quality of teaching and learning practices. The focal point of this review is threefold: to ascertain the focus of learning analytics research in online learning environments and its significant results, to identify the types of student-related analytics data in online learning environments and the issues related to analytics data, and to inform about the extent of interventions that have been applied in the learning analytics context, if any. Four procedures suggested by PRISMA were applied when conducting a systematic review. A total of 34 articles were chosen according to the review selection guideline by searching through online databases, including ACM Digital Library, LearnTechLib, ERIC, International Forum of Educational Technology & Society, ScienceDirect, Web of Science, Scopus, Society for Learning Analytics Research, and SpringerLink. The search keywords used were learning analytics data and online learning, learning analytics data and learning analytics intervention. The results were analysed based on the three focuses mentioned earlier and were interpreted accordingly. It is found that the applied focus of learning analytics is more oriented towards monitoring/analysis and prediction/intervention, and the commonly used types of student-analytics data are learning behaviour data and learning level data. Additionally, the intervention is still in the process of being developed. Lastly, the directions for future studies and limitations are also provided so as to further develop this emerging research area, and the findings can be a good reference point for other researchers.

Keywords Learning analytics · Online learning environment · Student-related analytics data · Intervention · Educational data mining

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1 Introduction

Learning analytics (LA) have been applied in many studies related to the educational technology field. This is due to the potential it holds in uncovering meaningful information on student behaviour in online learning environments, which in turn, reflects the influence of technology and learning design on the teaching and learning process. LA is used for the detection of problematic learning behaviours and affective states in students in online learning (Verbert et al., 2012; Kew & Tasir, 2017a, 2017b). One of the main functions of LA is to monitor the learning performance of students and identify potential learning problems early through logfile analysis so that interventions can be provided to recognize and support students at risk (EDUCAUSE 2010; Johnson et al., 2011). These logfiles in Learning Management System are generated by students when interacting in online activities, and with the use of LA, more hidden information about students in online learning can be discovered. Hence, it plays a significant role in online learning in order to identify problems with online learning and improve the learning environment. However, it is still a relatively new research area. During the 1st International Conference on Learning Analytics and Knowledge (LAK'11), LA was defined as “*the measurement, collection, analysis and reporting of data about learners and their contexts, for purposes of understanding and optimizing learning and environments in which it occurs*” (Long & Siemens, 2011, p.34). In addition, Cooper (2012, p.3) described analytics as “*the process of developing actionable insights through problem definition and the application of statistical models and analysis against existing and/or simulated future data*”. Although these definitions differ to some extent, they basically highlight the transactional information of particular data where learning progress, achievement, engagement, and other different aspects of learning can be measured, and the useful outcomes are predicted. Besides predicting, LA also allows the provision of dynamic feedback for a personalized learning experience and better intervention, since the action is the major aim of the analytics process.

The earliest analytics studies were applied in giant firms like Amazon, Google, and so on to help them make sense of their customers' data for the purpose of generating a higher business revenue. The analytics applied to these firms, however, require a complex technical process, and fully adopting the exact process in the educational field might not be applicable to academia (Gunn, 2014). The big data and analytics progressively have more implications in the education aspect (Seufert et al., 2019). This is because academicians are mostly interested in knowing and understanding meaningful traces of students' activity in online environments that can support the learning processes. Within this scope of interest, a great number of studies related to tracking and analysing the data from the online environment logs have started to emerge (Goldstein & Katz, 2005; Howell et al., 2018a, 2018b). The same is true for the development of dashboard applications that can visualize LA results and intervene automatically in the learning process whenever necessary. Some analytics dashboards are very system specific, while others are developed to be used across learning platforms (Arnold & Pistilli, 2012; Ferguson, 2012; Gunn, 2014; Norris & Baer, 2013).

While research on LA has continued to provide the basis for the field (West et al., 2018), it is often difficult to understand and interpret the bulk of the data collected, and not much is known about how students play a part in providing as well as attending to the messages of others in an online discussion environment (Wise et al., 2013). Furthermore, the designs of developed dashboards are far less effective and frequently lead to misinterpretation (Gasevic et al., 2015). Often, certain representations of student activity in

an online environment cannot be captured and presented efficiently (Corrin & de Barba, 2014). The types of data collected sometimes do not reflect clearly the pedagogical objectives that should be attained. When conducting research on LA, it is always important to link the pedagogical objectives with the types of data captured and assessed so as to provide data that are accessible to academic staff; thus, they are able to understand holistically about student interactions in online learning environments. Hence, this review article aims to offer the reader crucial details regarding the practical usage of LA techniques in online learning environments in order to improve the quality of teaching and learning practices. The focal point of this review is threefold: (1) to ascertain about the focus of LA research in online learning environments and its significant results, (2) to identify types of student-related analytics data in online learning environments and the issues related to analytics data, and (3) to inform about the extent of interventions that have been applied so far in online learning environments, if any.

2 Literature Review on Learning Analytics

LA is a relatively new field that has developed to assist in teaching and learning practices. Van Harmelen and Workman (2012) summarized the main promising uses of LA in the educational field, which are to identify at-risk students and provide them with positive interventions for retention improvement; to give recommendations to students regarding reading materials and learning tasks; to identify the demand for, and assess the outcomes of, pedagogic enhancements; to modify course offerings; to identify the teachers who perform well or who require assistance with their teaching methods; and to aid in the student recruitment process.

Chatti et al. (2012) also proposed a reference model for LA based on four dimensions, specifically, (i) what (e.g., data, environments and context), (ii) why (e.g., objectives), (iii) how (e.g., techniques/methods) and (iv) who (e.g., stakeholders), which has provided a clear overview of LA and its relevant concepts. In particular, the objectives of LA should be highlighted, as it can help stakeholders make effective decisions on teaching according to their learning focus. These LA objectives are listed in Table 1. Chatti et al. also reported that intelligent tutoring and adaptation was identified as the most frequently applied focus on LA objectives, followed by monitoring and analysis, assessment and feedback, prediction and intervention, and lastly, personalization and recommendation; nonetheless, no reflection of these was found in the research.

Moreover, for an in-depth understanding of how LA can maximize teaching and learning practices, the types of students' learning data play a role to help educators determine what data generated by students should be examined and analysed. In the research conducted by Wu et al. (2015), learners' learning status data are classified into four dimensions, as seen in Table 2.

The first type of data is learning behaviour data, where the data are tracked from between learners and the learning environment. The second type is learning emotional data in which the data are collected from the psychology and physiology of the learners in their online learning. Learning network data are the third type, where the data are harvested from the interaction of the learners with others in the online learning. Lastly, learning level data are defined as the data collected from the examination or the results achieved by the students.

Table 1 Objectives of LA in Learning Analytics Reference Model (Chatti et al., 2012)

Objectives	Descriptions
Monitoring/analysis	Tracking students' activities and generating reports to support decision making. Also related to the evaluation of the learning process by the teacher to improve the learning environment. Examining and analysing how the students use a learning system can assist teachers in detecting patterns and making decisions on the future design of learning activities
Prediction/intervention	Prediction is concerned with developing a model to predict the knowledge and future performance of students, which can then be used to develop an intervention, by suggesting actions that should be taken to help students who may need additional assistance
Tutoring/mentoring	In tutoring processes, the focus is on the teaching process (control of tutor). The focus of mentoring processes, meanwhile, is on the learning process of students
Assessment/feedback	It is concerned with providing the feedback to both students and teachers based on data about the students' interests and the learning context, supporting the assessment of the improved usefulness and efficiency of the learning process
Adaptation	Adaptation is triggered by the teacher/tutoring system or educational institution. It is concerned with guiding the students to perform the next move by adaptively organizing and establishing instructional activities and learning resources based on the needs of individual students
Personalization/recommendation	Personalization is concerned with assisting students to decide on their own learning and to continuously shape their own PLEs to achieve their learning goals. Meanwhile, the recommender system plays a role in fostering self-directed learning by recommending explicit and tacit knowledge nodes based on the preferences and activities of other learners with similar preferences
Reflection	It is concerned with promoting reflection from data compared within the same course, or across classes or institutions to draw conclusions on the efficiency of their learning or teaching practices

Furthermore, data science techniques have gradually become more visible in different fields, which can assist instructors and different stakeholders to make evidence-based decisions. One of the examples is that it is used to analyse the information derived from student interactions with games to enhance their interest in learning. In this aspect, different systematic reviews on LA have been conducted. For instances, Alonso-Fernández et al. (2019) had conducted a systematic literature review on applications of data science to game LA data and LA data from serious games. The results indicated that LA can be used to analyse the data to precisely predict the impact from an educational perspective. Similarly, another systematic review paper was conducted by Baiocchi (2019) who reviewed on the LA models and methods which are implemented to enhance the education on youth character, while Mangaroska and Giannakos (2018) carried out a systematic review on the empirical evidence on LA for learning design. Kew and Tasir (2017a, 2017b), on the other hand, reviewed identifying at-risk students in online learning by analysing learning behaviour through LA. Correspondingly, Schwendimann (2016) and Matcha et al (2019) carried out a systematic literature review of LA dashboards, while Wong (2020) and Kew and Tasir (2017a, 2017b) also conducted a systematic review on LA intervention. Nonetheless, among these review studies, limited studies were conducted to examine student-related

Table 2 Students' learning status data (Wu et al., 2015)

Students' learning status data	Description	Examples
Learning behaviour data	Data among learners and learning platform or learning resources	Log-in time, study time, number of clicks
Learning emotional data	Learners' psychological and physiological data during learning process	Post (feeling, emotion, expression etc.)
Learning network data	Learners' interactive data including network relationship and discourse content with others	Reply, post
Learning level data	Learners' exam or program scores reflecting learners' learning status directly	Grade

analytics data in an online learning environment. Therefore, this research aims to shed light on this aspect and make a contribution to the LA field by providing more insightful viewpoints on LA.

3 Research Questions

The following addresses the three research questions that motivated the review process:

Research Question 1: *What are the focuses (objectives) of learning analytics that researchers have applied so far in online learning environments, and what are the significant results?*

Research Question 2: *What are the types of student-related analytics data in online learning environments and the issues related to analytics data?*

Research Question 3: *To what extent have learning analytics interventions been applied, so far, in online learning environments, if any?*

4 Research Methodology

The systematic literature review was selected as a method to identify and review the practical usage of LA techniques in online learning environments. Based on the methodology employed, it is appropriate to classify this article as a systematic review article since it follows the guidelines proposed by and the four-phase flow diagram of PRISMA (Moher et al., 2009), as shown in Fig. 1. The guidelines proposed have been commonly and extensively utilized by a range of previous studies, showing it is an effective method for reviewing the articles. As suggested by Moher et al., four procedures/stages have been applied when conducting the review: (1) identification, (2) screening, (3) eligibility, and (4) included.

For the first procedure, the literature (from journal and conference articles) was gathered by searching through several well-known online databases and publishers for academic research, including ACM Digital Library, LearnTechLib, ERIC, International Forum of Educational Technology & Society, ScienceDirect, Web of Science, Scopus, Society for Learning Analytics Research (SoLAR) and SpringerLink. The search keywords used were learning analytics data and online learning, learning analytics data and learning analytics intervention. The targeted period of the article search was restricted from 2012 to 2020, in which LA bodies of knowledge have grown in number. The article searching process is until April 2020. The search process resulted in a total of 1,198 articles being found. Next, a screening process of the title and abstract was carried out. This resulted in the elimination of 786 articles.

After that, from the remaining 412 articles, further analysis was carried out to select articles that met the following criteria: (1) empirical articles associated with LA tools and student-related analytics data, (2) empirical articles related to LA intervention, and (3) empirical articles on LA that have a clear presentation of methodology, settings, findings, discussion, and interpretation of results. For this review article, LA intervention is defined as the “*frame of activity through which analytic tools, data, and reports are taken up and used*”, that articulates with the rest of the teaching and learning practices taking place (Wise, 2014, p.203). Based on the definition, we decided

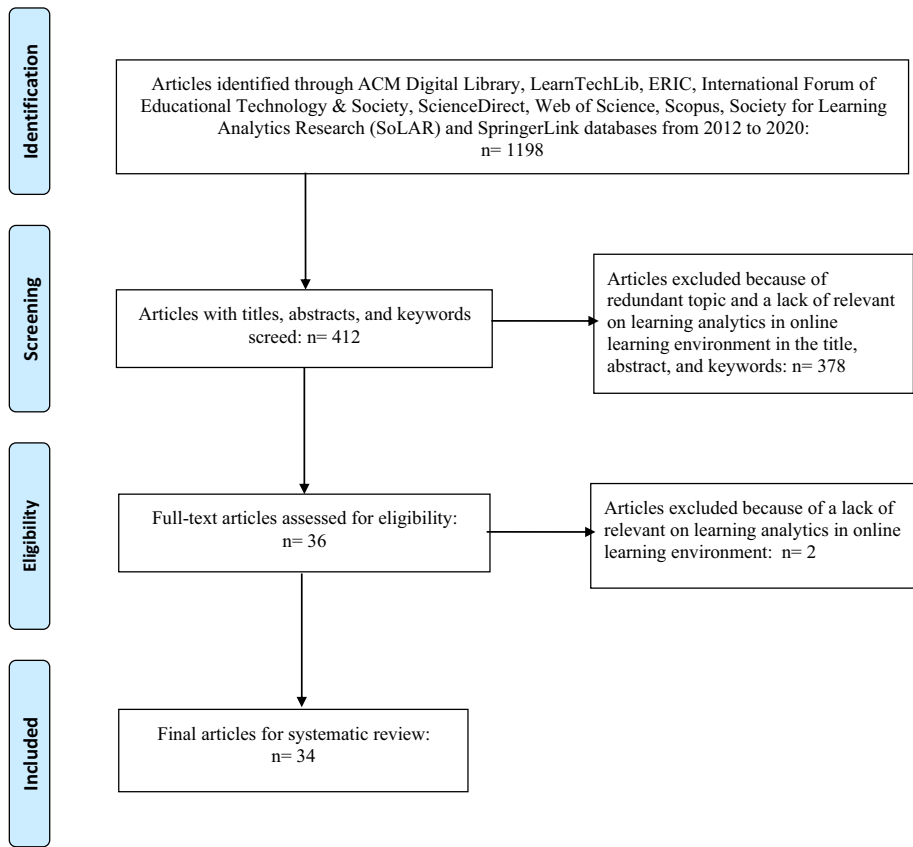


Fig. 1 Flowchart outlining the protocol adopted in this systematic review based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Four-Phase Flow Diagram (Moher et al., 2009)

to exclude articles related to two well-known projects on analytics, namely, Course Signals and the PAR framework. Although these two projects are related to LA intervention, the focus is more on academic analytics compared to LA as defined by Wise (2014). After eliminating articles which did not fulfil the above-mentioned criteria, a total of 34 articles were selected and analysed further. Next, the selected articles were analysed and classified according to the focus of LA research, types of student-related analytics data, and LA intervention. The focuses of LA research were classified according to the Learning Analytics Reference Model (see Table 1) proposed by Chatti et al. (2012), whilst types of student-related analytics data were categorized based on the Learners' Learning Status Data (see Table 2), as proposed by Wu et al. (2015). The findings were then reported using the descriptive analysis technique.

Finally, the presentation of the review based on the research questions was done. A critical review was synthesized by evaluating the empirical findings found in the 34 articles.

5 Results

This section is related to the findings of the review conducted on the selected 34 articles. All of the articles are based on empirical studies, which used either an exploratory, experimental, or evaluation research strategy. In terms of the learning settings of the studies (see Table 3), most studies are carried out in Learning Management System ($n=20$). Other preferred learning settings are Analytics Tool ($n=1$); Computer-based Environment ($n=5$), Virtual Learning Environment ($n=3$), and Web-based Environment ($n=5$).

An interesting question regarding LA is about the area of concern for it to benefit education. It is crucial to conduct LA practices with the focus (objective) in mind. The article classification according to the focuses of LA is tabulated in Table 4. As shown in the table, the primary focus in LA is related to monitoring/analysis (48.33%), where the majority of the studies reviewed here had investigated this focus of LA. The next focus is on prediction/intervention (35%), followed by assessment/feedback (10%), reflection (3.33%), adaptation (1.67%), and personalization/recommendation (1.67%). There is no study with a focus on tutoring/mentoring, at least for the scope of this review.

Another important aspect of LA is the types of student-related analytics data. It is important to be explicit about the types of data since the true potential of LA in teaching and learning is highly dependent on the accuracy of the data types that are chosen and analysed.

Based on Table 5, it can be seen that types of student-related analytics data mostly fall into the learning behaviour data category (37.04%), followed by learning level data (30.86%), learning network data (24.69%), and learning emotional data (7.41%). The authors have gathered data from the log files of a specific learning environment.

Table 6 illustrates that learning behaviour data are found most frequently in the focus of monitoring/analysis, followed by learning level data. Learning behaviour data is also found most frequently in other focuses (e.g., prediction/intervention, assessment/feedback,

Table 3 Classification of articles according to the learning settings

Learning setting	Authors and Year
Learning management system	Jo et al. (2015); Joksimovic et al. (2015); Lowes et al. (2015); Lotsari et al. (2014); Ma et al. (2015); Lim et al. (2019); Tempelaar et al. (2015); RUIPÉREZ-Valiente et al. (2015); Fidalgo-Blanco et al. (2015); Aljohani et al. (2019); Gašević et al. (2016); Kim et al. (2016); Colthorpe et al. (2015); Tempelaar et al. (2017); Lu et al. (2018); Zhang et al. (2018); Choi et al. (2018); Chen et al. (2018); Ibañez, Villalonga and Nuere (2019); Martin and Whitmer (2016)
Analytics tool	Haya et al. (2015)
Computer-based environment	Hershkovitz et al. (2013); Ji et al. (2015); Xing et al. (2015); Van Leeuwen et al. (2015); Howell et al., (2018a, 2018b)
Virtual-learning environment	Kennedy et al. (2013); Agudo-Peregrina et al. (2014); de Quincey et al. (2019)
Web-based environment	Wise et al. (2014); Kim et al. (2018); Aguiar et al. (2014); Lin and Hwang (2018); Prasad et al. (2016)

LMS—centralized educational system (e.g. Moodle, Blackboard, etc.); Analytics Tool—special environment designed for the needs of the study; Computer-based environment—environment that includes some type of computer technology (e.g. CSCL, desktop application); Virtual learning environment—a learning system with a virtual resource room feature; Web-based environment—web-based environment except LMS, VLE, MOOC

Table 4 Classification of articles according to the focuses of learning analytics

Focus	Authors and Year	Count	Percentage (%)
Monitoring/analysis	Haya et al. (2015); Hershkovitz et al. (2013); Ji et al. (2015); Jo et al. (2015); Joksimovic et al. (2015); Kennedy et al. (2013); Lotsari et al. (2014); Lowes et al. (2015); Ma et al. (2015); Wise et al. (2014); Xing et al. (2015); Agudo-Peregrina et al. (2014); Van Leeuwen et al. (2015); Kim et al. (2018); Tempelaar et al. (2015); Ruipérez-Valiente et al. (2015); Fidalgo-Blanco et al. (2015); Aljohani et al. (2019); Gašević et al. (2016); Kim et al. (2016); Aguiar et al. (2014); Colthorpe et al. (2015); de Quincey et al. (2019); Lin and Hwang (2018); Lu et al. (2018); Zhang et al. (2018); Prasad et al. (2016); Chen et al. (2018); Ibañez, Villalonga & Nuere (2019)	29	48.33
Prediction/intervention	Hershkovitz et al. (2013); Ji et al. (2015); Jo et al. (2015); Kennedy et al. (2013); Lotsari et al. (2014); Lowes et al. (2015); Ma et al. (2015); Wise et al. (2014); Xing et al. (2015); Lim et al. (2019); Van Leeuwen et al. (2015); Kim et al. (2018); Tempelaar et al. (2015); Gašević et al. (2016); Kim et al. (2016); Aguiar et al. (2014); Colthorpe et al. (2015); de Quincey et al. (2019); Tempelaar et al. (2017); Lu et al. (2018); Choi et al. (2018)	21	35.00
Tutoring/mentoring	-	0	0.00
Assessment/feedback	Kennedy et al. (2013); Lim et al. (2019); Tempelaar et al. (2015); Aljohani et al. (2019); Howell et al., (2018a, 2018b); Zhang et al. (2018)	6	10.00
Adaptation	Martin and Whitmer (2016)	1	1.67
Personalization/recommendation	Ruipérez-Valiente et al. (2015)	1	1.67
Reflection	Wise et al. (2014); Chen et al. (2018)	2	3.33
Total		60	100

Table 5 Classification of articles according to the types of student-related analytics data

Types	Authors and Year	Count	Percentage (%)
Learning behaviour data	Haya et al. (2015); Hershkovitz et al. (2013); Ji et al. (2015); Jo et al. (2015); Joksimovic et al. (2015); Kennedy et al. (2013); Lowes et al. (2015); Ma et al. (2015); Wise et al. (2015); Xing et al. (2014); Xing et al. (2015); Agudo-Peregrina et al. (2014); Lim et al. (2019); Van Leeuwen et al. (2015); Kim et al. (2018); Tempelaar et al. (2015); Ruipérez-Valiente et al. (2015); Aljohani et al. (2019); Gašević et al. (2016); Kim et al. (2016); Aguiar et al. (2014); Colthorpe et al. (2015); de Quincey et al. (2019); Tempelaar et al. (2017); Lin and Hwang (2018); Lu et al. (2018); Zhang et al. (2018); Prasad et al. (2016); Chen et al. (2018); Ibañez, Villalonga & Nuere (2019); Martin and Whitmer (2016)	30	37.04
Learning emotional data	Hershkovitz et al. (2013); Wise et al. (2014); Tempelaar et al. (2015); Ruipérez-Valiente et al. (2015); Howell et al., (2018a, 2018b); Tempelaar et al. (2017)	6	7.41
Learning network data	Haya et al. (2015); Hershkovitz et al. (2013); Ji et al. (2015); Lotsari et al. (2014); Lowes et al. (2015); Ma et al. (2015); Wise et al. (2014); Xing et al. (2015); Agudo-Peregrina et al. (2014); Van Leeuwen et al. (2015); Kim et al. (2018); Fidalgo-Blanco et al. (2015); Howell et al., (2018a, 2018b); Gašević et al. (2016); Kim et al. (2016); Lin and Hwang (2018); Lu et al. (2018); Zhang et al. (2018); Chen et al. (2018); Ibañez, Villalonga & Nuere (2019)	20	24.69
Learning level data	Jo et al. (2015); Joksimovic et al. (2015); Lotsari et al. (2014); Lowes et al. (2015); Xing et al. (2015); Agudo-Peregrina et al. (2014); Lim et al. (2019); Van Leeuwen et al. (2015); Kim et al. (2018); Tempelaar et al. (2015); Ruipérez-Valiente et al. (2015); Fidalgo-Blanco et al. (2015); Aljohani et al. (2019); Howell et al., (2018a, 2018b); Gašević et al. (2016); Kim et al. (2016); Aguiar et al. (2014); Colthorpe et al. (2015); de Quincey et al. (2019); Tempelaar et al. (2017); Lin and Hwang (2018); Lu et al. (2018); Zhang et al. (2018); Choi et al. (2018); Martin and Whitmer (2016)	25	30.86
Total		81	100

Table 6 Classification of articles according to the focuses of learning analytics versus the types of student-related analytics data

Focus	Types of student-related analytics data (number)			
	Learning behaviour data	Learning emotional data	Learning network data	Learning level data
Monitoring/analysis	28*	4	19	21
Prediction/intervention	18*	4	11	17
Assessment/feedback	5*	2	2	4
Reflection	2*	1	2*	–
Recommendation	1*	1*	–	1*
Adaption	1*	–	–	1*
Total	145			

*The highest number

reflection, recommendation, and adaption). These results help to reveal which types of student-related analytic data are most used in the focuses of LA.

Different kinds of learning settings have various functions and impacts of their own on teaching and learning practices. Table 7 shows that learning behaviour data and learning level data are the two types of student-related analytics data found mostly in LMS. Learning behaviour data was also found mostly in the virtual-learning environment and the web-based environment while the type of data used most in the computer-based environment is learning network data. On the other hand, in the analytics tool, learning behaviour data and learning network data are the two types of data that are being used most.

6 Discussion

In this section, we present the discussions based on the stated research questions.

Table 7 Classification of articles according to the learning setting versus types of student-related analytics data

Learning setting	Types of student-related analytics data (number)			
	Learning behaviour data	Learning emotional data	Learning network data	Learning level data
LMS	17*	3	9	17*
Analytics tool	1*	–	1*	–
Computer-based Environment	4	2	5*	3
Virtual-learning Environment	3*	–	1	2
Web-based environment	5*	1	3	3
Total	80			

*The highest number

RQ1: What are the focuses (objectives) of learning analytics do researchers applied so far in the online learning environment and its significant results?

In general, the most visible focuses of LA practices are on monitoring/analysis and prediction/intervention, and less attention was given to assessment/feedback, reflection, tutoring/mentoring, adaptation, and personalization/recommendation. However, intelligent tutoring and adaptation was found as the most frequently applied focus on LA objectives, followed by monitoring and analysis in the study conducted by Chatti et al. (2012). The shift in the trend is probably due to the growing interest in addressing how LA really functions in leveraging the effectiveness of the teaching and learning process. In other words, the focus should be on understanding the underlying pedagogy of the teaching and learning process rather than on the technology per se. As we know, adaptation is concerned more with data and technology. Being adaptive is fundamental to lifelong learning for individuals to be able to learn, unlearn, and relearn when new technologies change old tasks and produce new tasks (Chai et al., 2015). The adaptive system/program monitors the behaviour of a student while he/she is engaged with the system/program, and the learning experience is changed and improved over time based on the student's performance. For adaptive learning to function successfully, substantial preparation of the content is required to assist with the variation of learning pathways established by each student (Hoppes, 2015). To prepare the content, the faculty needs first to understand the underlying pedagogy which has caused the individual to behave differently in online learning. Due to this, increasingly, research on LA has started to focus on analysing students' online learning data to better understand and support the learning process. More teaching strategies can be used effectively based on these online learning data analyses in order to meet the demands of students in their learning process.

A common task in monitoring/analysis is to track students' online learning activities and produce reports for teachers and institutions so they can make a more informed decision regarding student performance and learning. Some of the studies that were reviewed under this category have revealed very important results which can benefit LA practice. Haya et al. (2015), for instance, stressed the importance of multiple levels of analysis in order to have deeper insights about the collaborative process. Meanwhile, Ji et al. (2015) stated that the difference between active and inactive students in relation to participation is due to the differences in usage in classes, which depended on the requirements of teachers. Next, Joksimovic et al. (2015) revealed that communications between students and instructors, as well as among students, would not be the main focus in foundation courses since the contents of such courses are mostly about basic knowledge, and students can handle such knowledge on their own. In contrast, the increased level of interaction between peers can be seen in the elective courses due to the close relevance to their final master's research. The core courses, on the other hand, showed an increase in communication between instructor and students because of the complexity of the course content, which requires extensive support from the instructor. This implies that the design of learning in an online environment should not be the same for every type of course level. Besides that, an LA approach was used by Lin and Hwang (2018) to analyse the factors affecting the students' oral presentation outcomes. In Ma's (2015) study, the results revealed that an instructor's course preparation is significantly and positively related to the students' viewing activities, while an instructor's guidance and assistance has a significant impact on the students' completion of the learning tasks. Correspondingly, Zhang et al. (2018) also reported that students' login behaviours, resource utilization, interactive behaviours, etc. can affect students' learning process and results of online learning. Kim et al. (2018) also

applied LA to identify student profiles by analysing log data associated to self-regulated learning in an online course, including online study time, log-in numbers, test scores, etc. Three other studies (Jo et al., 2015; Lowes et al., 2015; Xing et al., 2015) also highlighted that LA practice should not be data driven only but should be theory driven as well. This is because the theoretical underpinning is able to address a student's particular competency in the multifaceted complexities of online learning.

Prediction/intervention is also a key objective in LA research. Prediction is concerned with developing a model to predict knowledge and future performance, and the developed model can be used for intervention purposes, which can help struggling students to succeed in a course. Regarding intervention, only a few studies focused on this aspect, including Choi et al.'s (2018) study, Wise et al.'s (2014) study and so on. The prediction models are often developed through data mining techniques (e.g. Clustering, Classification, Association Rule), or statistical analyses (e.g. Regression, Structural Equation Modeling, SEM), or both. Some of the interesting models that have been created are related to student carelessness in learning (Hershkovitz et al., 2013), the learner activity model in a collaborative learning system (Ji et al., 2015), the proxy variables for adult learners' time management strategies (Jo et al., 2015), a model for learning surgical techniques through simulation (Kennedy et al., 2013), a method of student profiling in online discussions (Lotsari et al., 2014), a model of the relationship between engagement and final grade (Lowes et al., 2015), a teaching and learning interaction activities model (Ma et al., 2015), and a participation-based student final performance prediction model (Gašević et al., 2016; Xing et al., 2015). Some authors (Lotsari et al., 2014; Lowes et al., 2015) have used only one predictive algorithm in constructing the prediction model, whereas others (Kennedy et al., 2013; Xing et al., 2015) have opted to use more than one predictive algorithm. Xing et al. (2015) stated that algorithm selection is the key to developing an effective performance prediction model, and while some modelling algorithms (e.g., artificial neural networks) have a good prediction rate in other disciplines, those algorithms may not necessarily function equally well when addressing educational problems in each case. This is due to the limitations in the availability of training data. Lu et al. (2018) predicted the final academic performance of students by using regression, and the findings revealed that the final academic performance of students could be predicted after one-third of the semester. Ji et al. (2015), meanwhile, claimed that a model can be developed regardless of data size by carefully selecting features that illustrate learners' activities from log data, and the developed model can be tested later for validation purposes. As the algorithms used become increasingly complex, it will cause some difficulties for teachers and students in interpreting the reports during the implementation (Kennedy et al., 2013; Xing et al., 2015). Therefore, researchers need to provide guidelines on how to interpret the data so as to avoid negative understanding and conclusions.

Intervention, assessment/ feedback, tutoring / mentoring, adaptation, and personalization/ recommendation are types of research which are oriented to the application of data. From Table 4, it can be seen that not many studies have been conducted under these categories. This is because these types of study are complex to execute (in terms of data management and appropriate visualization) and require accurate data profiling to function effectively. The data also need to be processed and presented to the stakeholders (e.g., student, teacher, institution) to support decision making. Knowing the stakeholder's needs precisely will directly influence the development of the analytics process. As for the suggestion, to optimize the teaching and learning process, future LA practices should place the student and teacher as the most important stakeholders. This can be done by offering ways that allow teacher and student to reflect on and draw conclusions from the effectiveness of

their teaching method and learning behaviour, and give appropriate feedback as a means to intervene in the learning process. Therefore, more research related to feedback, adaptation, intervention, and reflection needs to be conducted in the future, and the challenges that surround the process need to be addressed properly.

LA technologies in terms of the tools/environments that have been used in the review studies can be classified into five main categories (LMS, Analytics tool, Computer-supported environment, Virtual learning environment and Web-based environment), with LMS topping the list since the trend for LA research in the past few years has focused on centralized learning systems. Basically, research that is related to monitoring/analysis is suitable for any type of learning environment as long as that environment is able to track the log files of teacher and student activities when using the system, and can offer a simple way of monitoring and assessing performance. However, a suitable learning environment is needed if the aim is to attain a particular pedagogical goal. For instance, Kennedy et al. (2013) conducted research related to providing timely feedback to students when learning surgical techniques through a virtual learning environment since the simulation feature is not available in the LMS environment. Quincey et al. (2019) used the visualization form for student-centred design. Ibañez, Villalonga and Nuere (2019) analysed the data from LMS, such as user behaviour, user activity, etc. to explore their students' activity. Haya et al. (2015), meanwhile, used the Social Learning Analytics Toolkit combined with social network analysis and content analysis to assess the collaboration process for a video production task. Furthermore, Wise et al. (2014) utilized a web-based environment named Starburst to embed analytics for intervention purposes, which cannot be done through the LMS environment. We also noticed that, in this review, there is no study which links Mobility environment. This environment has one thing in common, that is, visualization (e.g., videos, diagrams, and tables), and it should be noted that visualization is strongly related to LA research interests. Therefore, it will be interesting if future LA research could put heavier emphasis on investigating the essential learning interaction through this suggested environment so as to fully appreciate the role they play in presenting education.

RQ2: What types of student-related analytics data in an online learning environment and the issues are related to analytics data?

The most investigated student-related analytics data is learning behaviour data, whereas the least examined student-related analytics data is learning emotional data. This finding is expected, as learning behaviour data is suitable for providing insights into student participation in online courses, for example, the number of logins, the number of pages accessed, the time spent in the system, and so on. Plus, it is much easier to gather this type of data through the LMS used. As researchers have investigated the behaviour data, which is linked to success, the behaviour chosen has varied since it depends on what behaviours the researchers believe are important in their research context as well as what data are available in the learning environment setting. For instance, Lowes et al. (2015) investigated how online participation behaviours are related to course outcomes. They found that all three frequency behaviours (number of days accessed system, number of times logged into system, total spent logged) successfully predicted the course grade. Prasad, Rajneel and Usagawa (2016) tracked learners' online and offline interactions with open textbooks such as total views of chapter, etc. On the other hand, Kennedy et al. (2013) assessed 48 behaviours generated by the 3D environment and discovered that these behaviour variables can provide meaningful feedback in surgical education and training. While learning behavioural data continue to provide insights into what is going on inside courses, there is one

major limitation to the use of this behaviour data: this type of learning data allows only a partial/surface inspection of engagement/interaction, and it is not suitable to be used if a researcher wants to investigate a deeper level of engagement/interaction (for instance, the qualitative aspects of instructor-students and student-student engagement/interaction) and how students/instructor attained a certain level of engagement. This limitation has driven other researchers (Haya et al., 2015; Hershkovitz et al., 2013; Ji et al., 2015; Lotsari et al., 2014; Lowes et al., 2015; Ma et al., 2015; Wise et al., 2014; Xing et al., 2015; Zhang et al., 2018; Ibañez, Villalonga & Nuere, 2019) to focus not only on the behaviour data but on the learning network data as well since this type of data is critical in explaining the complexities of the learning process which takes place in an online setting.

Among the outcomes that have been achieved by analysing learning network data are (1) revealing how students contribute to and attend to the messages of others, (2) facilitating productive decision making and enabling reflection based on the current status of the discussion forum, and (3) implementing appropriate intervention for changes in teaching and learning pedagogical strategies. It should be noted that most of the studies reviewed here have considered some dimensions of network data like post forums, reply forums, and so on. Also, there is limited research that examines how the unstructured data (unfixed data), like pictures, videos, and so on, can influence the discourse content among students and instructor. Moreover, each study has used only one type of online learning environment and has not considered data that come from the outside learning environment, for instance, Facebook, Skype, blogs, and so on, which might have strengthened the learning network data. Therefore, we expect more detailed studies to consider the abovementioned aspect, as this is the most popular trend in LA. Further verification of the findings of the reviewed studies should also be conducted to ascertain the effectiveness of learning network data in teaching and learning and to evaluate the transferability of the research across different fields and settings.

From Table 5, it is also noticed that a large number of studies used learning level data (e.g., exam grade) since this is the fastest and simplest indicator/measure that one can refer to in order to know about the overview performance in learning. Most of the studies used this type of data to predict students' performance and to identify at-risk students. However, it can also be seen that not many studies have examined learning emotional data (e.g., post about feelings). For learning emotional data, more research needs to be conducted though it may be difficult to interpret this kind of data precisely and might require a significant amount of interpretation time, as the data are abstract in nature. This is because students' emotions could be one of the main reasons for dropout rates, and different students will experience different emotions and feelings at different times even in the same situation (Pekrun, 2014). Additionally, reducing learners' drop-out rates was the main objective of many education researchers and practitioners. Therefore, it is suggested that more attention should be paid to the research related to the psychological aspect of students, such as motivation, in order to enhance the teaching and learning success.

Furthermore, Table 6 shows learning behaviour data are mostly used in the focus of the monitoring/ analysis and the prediction and intervention to support the teaching and learning practices. From Table 7, it could also be seen that learning behaviour data and learning level data are the two types of data most used in LMS, whilst learning network data are used highly in computer-based environments. These results give us an overview of the types of student-related analytics data which are highly used in the focus of LA objectives and learning settings. From the reviewed articles, we became aware that identifying and collecting the right type of analytics data is fundamental for the LA process. Yet, big issues emerged in relation to analytics data, and these require some

consideration regarding LA practices. These include the following: (1) it is not fully understood how a huge student dataset can be managed and integrated, (2) it is not certain how LA results are disseminated through the student, faculty, and institution, and back again, and (3) it is not certain to what extent ethics should be considered in order to practise LA in higher learning institutions. Advances in computational technology in higher learning institutions have made possible the support of a large volume of students' data. However, these data are often stored separately, and higher education "*has traditionally been inefficient in its data use, often operating with substantial delays in analysing readily evident data and feedback*" (Long & Siemens, 2011, p.32).

To date, less is known about how the aggregation of data can present meaningful information about the teaching and learning process. It is beyond the scope of this paper to debate about an integrated or disintegrated system to practise the LA approach; however, there is a need for an integrated, synchronized, and holistic approach involving all stakeholders. The review of the articles somehow shows that, often, the LA findings are made available not to students and the institutions but to teachers only; nevertheless, it should be noticed that teachers are not the only actors that are responsible for students' success. This raises the point that the decision to act based on the LA findings should be made by other stakeholders who can contribute to and use the analyses; these include students, faculties, and institutions. By doing so, it may increase the understanding of student success, and appropriate actions can be tailored to achieve a higher value of teaching and learning for the institution. The ethical issues regarding the use of data for making decisions are a new subject for the educational sector. There is an obligation on the part of the institutions to formulate the codes of conduct, which detail the rights of each stakeholder and which cover informed consent, the purpose of data tracking, data ownership, and the extents of data usage (Land & Bayne, 2005) and to make it known if the data collected from students are used for other purposes. To sum up, when dealing with analytics data, it is important to remember to always "get the right data and get the data right".

RQ3: To what extent have learning analytics interventions been applied, so far, in online learning environments, if any?

According to Wise et al. (2014), the purpose of LA intervention is to serve as validation of the potential effectiveness of LA for influencing student activity in online discussions. Among the 34 articles reviewed here, only a few studies (Choi et al., 2018; de Quincey et al., 2019; Lim et al., 2019; Wise et al., 2014) focused on the intervention aspect, which studied the framing of the activity (for teachers and/or students) of interpreting and making decisions based on the analytics. Lim et al. (2019) provided an LA-based feedback intervention while de Quincey et al. (2019) provided intervention in the form of weekly personalised notifications to students in order to enhance their learning performance. Choi et al. (2018) designed a systematic proactive intervention for students (e.g., emails, phone calls, etc.), and the result showed that the number of interventions is positively associated with the efficacy. Wise et al. (2014) first overviewed the E-Listening research program and translated the work into analytics that students and instructors could use to reflect on their discussion participation. Findings from an initial implementation of the application indicated that the LA intervention supported changes in students' discussion participation. They further highlighted that another vital but less researched aspect is the model for intervention. Developing an intervention model that can increase learners' learning performance is significant in prompting learning and teaching and developing LA (Wu et al.,

2015). This is aligned with the findings from the systematic review by Wong (2020) about how more research on LA intervention is needed.

Further searching through the literature revealed that the current trend for LA intervention is basically about the construction of a model or framework to guide the execution of LA intervention. Wu et al. (2015) designed the intervention model and strategy based on the behaviour data of learners. The core of the intervention model is an intervention engine containing learners' learning status identification, intervention strategy matching and calculation, intervention application, and intervention effectiveness analysis. Kew and Tasir (2016) meanwhile proposed the LA intervention framework, which focuses on enhancing motivation, cognitive engagement, cognitive retention, and academic performance for students with different learning styles. Besides that, some other studies related to LA were also conducted to enhance teaching practices. For instance, Lonn et al. (2015) investigated students' motivation in the context of an LA intervention during a summer bridge program, and it was suggested that students' perceptions about their goal and performance are essentially paid attention while designing LA intervention. Wise et al. (2014) were also concerned with the application of LA in investigating the ways learners give and respond to peers' messages on online discussions, and an intervention was designed by referring to the principles proposed by Wise et al. (2013), which showed a positive result. Moreover, some successful institutions which have utilized LA tools for intervention purposes have confirmed the important role of intervention in higher education institutions; for example, Rio Salado Community College with Progress and Course Engagement, University of Michigan with E^2 Coach, Marist College with Open Academic Analytics Initiative, Northern Arizona University with Grade Performance Status, and Open University, UK which drives intervention at the student level.

For the intervention to be useful in guiding decision making by teachers and students, a proper intervention method should be framed. Besides the well-designed metrics, most teachers and students need guidelines and training on how to read and interpret the LA output, so they can understand what needs to change and how to address that change (Wise et al., 2014). Although current works on LA intervention help to provide a point of reference for us to expand the valuable attribute for conducting intervention in an online learning environment in the near future, we believe that one model/framework on intervention will not fit exactly with every mission in improving teaching and learning performance since online learning is very complex in nature. Therefore, more LA interventions should be developed, which was also proposed by Wong (2020) and Kew and Tasir (2017a, 2017b).

All in all, this section has discussed the findings and challenges in LA that need further review and research to examine and overcome them. In the future, the use of LA and the algorithms in the educational field will become increasingly complex, which means teachers will face some difficulties in interpreting the reports during the implementation. To resolve this, a clear guideline needs to be provided on how to interpret the data in order to get a more precise understanding. Besides that, it would be more challenging to fully understand about students from the psychological aspect in online learning. It is then proposed that future research should pay more attention to elements of this aspect, such as motivation, satisfaction, confidence, anxiety, etc., in order to enhance the success of teaching and learning. Additionally, different instructors and institutions will increasingly integrate different online settings; thus, more research related to different objectives and different types of student-related analytics data need to be carried out to tackle this challenge. Also, as mentioned earlier, some issues need to be resolved in the future as well, such as (1) it is not fully understood how a huge student dataset can be managed and integrated, (2) it

is not certain how LA results are disseminated through the student, faculty, and institution, and back again, and (3) it is not certain to what extent ethics should be considered in order to practice LA in higher learning institutions. Furthermore, as the amount of online learning will be growing rapidly, more LA interventions should be developed to assist students.

7 Conclusion and Implications

Among the aforementioned review studies on LA, the studies which examine student-related analytics data including the focus of LA research and types of student-related analytics data in the online learning environment are still vague. However, this review paper has contributed to knowledge of this aspect by providing an overview of LA. This review paper has described the efforts made to offer the reader the necessary details regarding the practical usage of LA techniques in an online learning environment, and the review has shown that (1) the applied focus of LA is more oriented towards monitoring/analysis and prediction/intervention, (2) the commonly used types of student-analytics data are learning behaviour data and learning level data, and (3) the intervention is still in the process of being developed where researchers are investing a great amount of effort in constructing models or frameworks to execute the intervention process. We are certain that the shift in focus will occur as this field evolves due to the emergence of new technologies which can be explored from the LA viewpoint. In addition, the challenges that have been stated need to be addressed in order to get credible results and to develop an understanding of how LA can be applied in its full complexity.

Besides that, this review paper generates some implications for practice. The findings showed that LA is greatly utilised by the researchers, and since the data trail from the online learning environment that is created by students when they engage with such a system can be retrieved and analysed to discover meaningful data about learners, it is believed that it can enhance the quality of online teaching and learning practices. For example, different types of data can be tracked, analysed, and interpreted in order to meet the different goals, including formulating different frameworks of learning behaviour of students and building fundamental knowledge of LA. In particular, Ifenthaler and Widanapathirana (2014) highlighted that theoretical concepts and empirical evidence need to be generated within the LA field. Additionally, from this study, it is found that most of LA focus on monitoring/ analysis, and the most commonly used type of data is learning behaviour data, showing that LA could be related to technology integration in online learning from behaviourist perspective that aims to understand students' learning behaviour in order to improve learning. This can then be used to formulate the framework or model. Also, it is vital to link with the pedagogical objectives with the types of data captured and assessed so as to provide data that are accessible to academic staff; thus, they are able to understand holistically about student interactions in online learning environments. To prepare the content, the instructors need first to have some idea of the underlying pedagogy which has had an impact on the individual so they behave differently in online learning. Due to this, increasingly, research on LA has started to focus on analysing students' online learning data to better understand and support the learning process. Therefore, this showed that the instructors should consistently apply LA as one of their teaching strategies to monitor the learning progress of students through learning behaviour data. On the other hand, learning level data focuses significantly on the performance of students in online learning, as a more effective learning

environment is designed to help students actively organize information by making the internal cognitive connections underpinning cognitivism learning theory (Koç, 2005). These can be taken into consideration and used as a reference by instructors for effective technology integration in online learning through LA, in particular when designing teaching materials or intervention for students in order to support the learning process.

Moreover, the findings from this review paper also informed the types of LA technologies in terms of the tools/environment which have contributed to the knowledge body of LA, particularly the methodology aspect. It is found that some areas are still under-researched, for example, LA tools, as there was only one article. Thus, more studies should be carried out in the future because it is an important area to be investigated. In particular, Haya et al. (2015) highlighted the importance of multiple levels of analysis to have a deeper understanding about the learning process. Additionally, learning and teaching processes are dynamic, where multimodal analysis is needed to help instructors understand their students' learning processes in online learning (Şahin and Yurdugül, 2020). This finding also encourages more researchers to develop more LA technologies to assist the instructors in online learning. Also, there is limited research that examines how the unstructured data, for example, graphics and videos, can influence the discourse content among students and their instructor. It is also noticed that limited studies have examined learning emotional data (e.g., posts on feelings), as the data are abstract in nature. Therefore, based on the findings of the review, instructors are encouraged to focus on these underinvestigated research areas, since the integration of LA in online teaching practice has a variety of useful functions. In particular, it can be used to discover the undesirable learning behaviours through monitoring or predicting and then to provide intervention to students based on the outcome of data analysis to solve online learning issues.

8 Limitations and Future Studies

While the result for this paper has contributed to the body of knowledge in LA and some useful suggestions for future studies have been highlighted in the discussion section, this paper is not without its limitations. One of the limitations is that only 34 articles from 2012–2020 were reviewed; thus, more systematic reviews on this topic are needed in the future. Some parts of these LA articles were not looked into, such as the research methodology part, including how these LA studies were conducted and how the authors visualized their LA findings. It is also suggested that future studies can analyse this part by referring to the five approaches proposed by Ferguson and Shum (2012): social network analytics, discourse analytics, content analytics, disposition analytics and context analytics. Additionally, the researchers can consider analysing multimodal multichannel data and LA algorithms and techniques. Other than that, the review on which role of stakeholders is important and information about who is affected in these LA studies have not been analysed. Therefore, these limitations should be taken into consideration in the future so that the potential use of LA can be maximized.

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